

extracted from it. A common way to trade this is to have the crude on one leg and gasoline and heating oil on the other, usually in a ratio of 3:2:1. You can essentially make up any sort of spread combination you like, but do ensure that the contracts have a real-world connection and reason why they move in similar ways. If you pick contracts that just happened to have similar return profiles for a while but could potentially decouple, you may end up taking a larger risk than you bargained for.

■ **Correlation Matrices, Position Sizing, and Risk**

If you are serious about the field of managed futures and trend following, this is an area in which you may need to do some hard work. After you have replicated and fully understood all other aspects of diversified trend following, this is likely to be the area where you need to spend the most time to perfect your strategy design.

If anyone asked me what the single biggest flaw is in the core strategy we have used up to this point, my answer would be that the strategy treats every position completely independently and does not take into account how related they are to each other. The position sizing and risk management are therefore done on a position level and not on a portfolio level. There are two main reasons why I have let this flaw remain in the core strategy. First, it works quite well without it, showing strong results at acceptable volatility and drawdowns. Second, the topic is complex enough to warrant another few hundred pages to treat properly.

Consider a situation where there has been one overwhelming theme dominating the real-world developments for some time. Perhaps there is a serious bear market in equities, primarily driven by negative European Union developments, as in 2011/2012. This may have caused most bond futures to trend up for a prolonged period of time while the dollar gained strength and commodities moved down. So we have a fully invested portfolio with long bonds, short equities, short currencies against the dollar, and short commodities. The question then is: do we have many different trades or one single bet on the same theme? It may be fine that we just have a concentrated bet, because that is how the strategy is designed and what happens in some markets, but the risk level will be very high and the more these positions go in our favour, the larger the difference will be in the risk level intended by our position-sizing formula and the actual risk level. With such